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LONG-TYPE KEY STRUCTURE

Abstract

A long-type key structure is provided. The structure of the long-type key structure is specially designed. Consequently, the vibration noise of the metal moving parts and the key pressing stroke delay can be effectively reduced, and the pressing feel can be enhanced. The long-type key structure includes a keycap, a base plate and a supporting element. The supporting element includes an outer scissor frame, at least two inner scissor frame and a metallic reinforcement plate. The metallic reinforcement plate is combined on the surface of the outer scissor frame.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to the structure of an input device, and more particularly to a key structure.

BACKGROUND OF THE INVENTION

[0002] In modern societies, electronic products become indispensable parts in human lives. The electronic products are applied in many sectors, including food, clothing, housing, transportation, education and entertainment. Generally, the electronic product is equipped with a keyboard for facilitating the user to input control signals.

[0003] Conventionally, a scissor-type supporting structure is one of the supporting elements for the key structures of the keyboard. The scissor-type supporting structure is made of a plastic material. When the scissor-type supporting structure is applied to a long-type key structure, the scissor-type supporting structure cannot provide sufficient structural strength for transmitting the pressing force because the plastic material has its strength limit. Consequently, when the corner of the keycap of the key structure is pressed by the user, the pressing feel is usually not satisfied.

[0004] In order to increase the structural strength of the supporting element, the existing technology also uses two metal brackets and a plastic scissor-type supporting structure to overcome the insufficient structural strength problem of the scissor-type supporting structure.

[0005] The additional metal brackets may significantly increase the structural strength. However, the addition of the metal brackets increases the manufacturing cost and the assembling cost of the overall key structure. Furthermore, when the metal brackets and similar moving parts are rotated or swung, the vibration noise is readily generated. In other words, the uses of the additional metal brackets cannot meet the needs of some consumers who pursue extremely quiet use while working.

[0006] In order to overcome the drawbacks of the conventional technologies, it is important to provide an improved long-type key structure for reducing the vibration noise when the metal moving parts are rotated or swung, reducing the key pressing stroke delay when the key structure is pressed, and enhancing the pressing feel.

SUMMARY OF THE INVENTION

[0007] An object of the present invention provides a long-type key structure for reducing the vibration noise when the metal moving parts are rotated or swung. In addition, when the long-type key structure is pressed, the key pressing stroke delay is reduced, and the pressing feel is enhanced.

[0008] In accordance with an aspect of the present invention, a long-type key structure is provided. The long-type key structure includes a keycap, a base plate and a supporting element. The supporting element includes an outer scissor frame, at least two inner scissor frames and a metallic reinforcement plate. The outer scissor frame includes a first end and a second end opposed to the first end. Each of the at least two inner scissor frames includes a third end and a fourth end opposed to the third end. The at least two inner scissor frames are pivotally coupled to the outer scissor frame and permitted to be swung relative to the outer scissor frame. The metallic reinforcement plate has a shape and a structure matching the outer scissor frame. The metallic reinforcement plate is combined on a surface of the outer scissor frame. The first end and the third end are connected with the keycap. The second end and the fourth end are connected with the base plate.

Consequently, the keycap is movable upwardly or downwardly relative to the base plate.

[0009] In an embodiment, a bottom surface of the keycap is provided with plural rotary connection parts and plural movable connection parts, a first rod part is formed on the first end, a third rod part is formed on the third end, the first rod part is movably connected with the plural movable connection parts, and the third rod part is rotatably connected with the plural rotary connection parts.

[0010] In an embodiment, the third rod part is rotatably connected with at least two rotary connection parts of the plural rotary connection parts.

[0011] In an embodiment, the base plate includes plural base connection parts, a second rod part is

formed on the second end, a fourth rod part is formed on the fourth end, and the second rod part and the fourth rod part are movably connected with the base connection parts.

[0012] In an embodiment, the long-type key structure further includes a metallic reinforcement bar. The metallic reinforcement bar is installed on a bottom surface of the keycap and circumferentially arranged beside a long side and a short side of the keycap.

[0013] In an embodiment, plural coupling parts are formed on the bottom surface of the keycap and arranged beside the long side and the short side of the keycap. The metallic reinforcement bar is coupled with the plural coupling parts.

[0014] In an embodiment, a raised platform structure is formed on the bottom surface of the keycap and arranged beside the short side of the keycap, and a portion of the metallic reinforcement bar beside the short side of the keycap is pushed by the raised platform structure. A distance between a top surface of the short side of the keycap and the metallic reinforcement bar is larger than a distance between the top surface of the long side of the keycap and the metallic reinforcement bar.

[0015] In an embodiment, a bent structure is formed on a periphery region of the metallic reinforcement plate, so that a weight of the metallic reinforcement plate is reduced.

[0016] From the above descriptions, the present invention provides a long-type key structure. Due to the combination of the outer scissor frame and the metallic reinforcement plate, the overall structural strength of the supporting element will be increased. When the key structure is pressed, the pressing force can be effectively transmitted through the supporting element, and the vibration noise generated when the metal moving parts are rotated or swung will be effectively reduced.

[0017] Furthermore, the third rod part of each inner scissor frame is connected with at least two rotary connection parts of the keycap. Consequently, the pressing feel of pressing the corner of the keycap can be effectively enhanced. Moreover, since the metallic reinforcement bars and the raised platform structures are installed on the long side and the short side of the bottom surface of the keycap, the pressing feel and key triggering sensitivity of pressing the key structure will be enhanced. In addition, the key pressing stroke delay can be effectively reduced.

[0018] The above objects and advantages of the present invention will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1A is a schematic perspective view illustrating a long-type key structure according to an embodiment of the present invention;

[0020] FIG. 1B is a schematic exploded view illustrating the long-type key structure according to the embodiment of the present invention;

[0021] FIG. 2 is a schematic perspective view illustrating the keycap of the long-type key structure according to the embodiment of the present invention and taken along a viewpoint;

[0022] FIG. 3 is a schematic perspective view illustrating the supporting element of the long-type key structure according to the embodiment of the present invention and taken along a viewpoint;

[0023] FIG. 4 is a schematic perspective view illustrating the base plate of the long-type key structure according to the embodiment of the present invention and taken along a viewpoint;

[0024] FIGS. 5A and 5B are schematic cross-sectional views illustrating the long-type key structure according to the embodiment of the present invention and taken along different viewpoints;

[0025] FIG. 6A is a schematic cutaway view illustrating the installation of the metallic reinforcement bar in the long-type key structure of the present invention;

[0026] FIG. 6B is a schematic side view illustrating the long side of the keycap in the long-type

key structure of the present invention;

[0027] FIG. 6C is a schematic side view illustrating the short side of the keycap in the long-type key structure of the present invention;

[0028] FIG. 7A is a schematic perspective view illustrating a portion of the structure of a first exemplary metallic reinforcement bar; and

[0029] FIG. 7B is a schematic perspective view illustrating a portion of the structure of a second exemplary metallic reinforcement bar.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0030] The present invention will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this invention are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed.

[0031] Please refer to FIG. 1A and FIG. 1B. FIG. 1A is a schematic perspective view illustrating a long-type key structure according to an embodiment of the present invention. FIG. 1B is a schematic exploded view illustrating the long-type key structure according to the embodiment of the present invention. In an embodiment, the long-type key structure **1** comprises a keycap **10**, two metallic reinforcement bars **20**, a supporting element **30** and a base plate **40**.

[0032] The supporting element **30** comprises an outer scissor frame **31**, two inner scissor frames **32** and a metallic reinforcement plate **33**. The inner scissor frames **32** and the outer scissor frame **31** are pivotally coupled to each other. Consequently, the inner scissor frames **32** can be swung relative to the outer scissor frame **31**. The metallic reinforcement plate **33** is correspondingly connected with the surface of the outer scissor frame **31**. For succinctness, the switch circuit board, the elastic element and the associated component over the base plate **40** are omitted.

[0033] FIG. 2 is a schematic perspective view illustrating the keycap of the long-type key structure according to the embodiment of the present invention and taken along a viewpoint. The keycap **10** has a top surface **11** and a bottom surface **12**. The bottom surface **12** of the keycap **10** is provided with plural rotary connection parts **121**, plural movable connection parts **122** and plural coupling parts **123** and **124**. For example, the plural rotary connection parts **121** include six rotary connection parts **121** in a linear arrangement, and the plural movable connection parts **122** include six movable connection parts **122** in a linear arrangement. The coupling parts **123** are located at the long sides of the keycap **10**. The coupling parts **124** are located at the short sides of the keycap **10**. The metallic reinforcement bars **20** shown in FIG. 1B are coupled by the coupling parts **123** and **124**. In an embodiment, the bottom surface **12** of the keycap **10** is further provided with raised platform structures **125** at the short sides of the keycap **10** adjacent to the corners of the coupling parts **124**.

[0034] FIG. 3 is a schematic perspective view illustrating the supporting element of the long-type key structure according to the embodiment of the present invention and taken along a viewpoint. The outer scissor frame **31** and the inner scissor frames **32** are made of non-metallic material (e.g., plastic material). The outer scissor frame **31** has a first end **311** and a second end **312**, which are opposed to each other. A first rod part **3111** is formed on the first end **311** of the outer scissor frame **31**. A second rod part **3121** is formed on the second end **312** of the outer scissor frame **31**. The inner scissor frame **32** has a third end **321** and a fourth end **322**, which are opposed to each other. A third rod part **3211** is formed on the third end **321** of the inner scissor frame **32**. A fourth rod part **3221** is formed on the fourth end **322** of the inner scissor frame **32**. In this embodiment, the total surface area of the outer scissor frame **31** is greater than the total surface area of the two inner scissor frames **32**. In addition, the outer scissor frame **31** comprises two mounting openings **313**. The shapes of the mounting openings **313** match the shapes of the corresponding inner scissor frames **32**. The inner scissor frames **32** are installed in the corresponding mounting openings **313** and pivotally coupled to the outer scissor frame **31**. Consequently, the inner scissor frames **32** can be swung relative to the outer scissor frame **31**.

[0035] The shape and the structure of the metallic reinforcement plate **33** match the shape and the structure of the outer scissor frame **31**. Since the outer scissor frame **31** has the large total surface area, the metallic reinforcement plate **33** can be combined on the surface of the outer scissor frame **31** by using a hot melt process or an over-molding process. Due to the combination between the metallic reinforcement plate **33** and the outer scissor frame **31**, the structural strength of the supporting element **30** is effectively enhanced. In this embodiment, the metallic reinforcement plate **33** is combined on the top surface of the outer scissor frame **31**. It is noted that numerous modifications and alterations may be made while retaining the teachings of the invention. For example, in a variant example, the metallic reinforcement plate **33** is combined on the bottom surface of the outer scissor frame **31**.

[0036] FIG. **4** is a schematic perspective view illustrating the base plate of the long-type key structure according to the embodiment of the present invention and taken along a viewpoint. The base plate **40** comprises plural base connection parts **41** and **42**.

[0037] Please refer to FIGS. **5A** and **5B**. FIGS. **5A** and **5B** are schematic cross-sectional views illustrating the long-type key structure according to the embodiment of the present invention and taken along different viewpoints.

[0038] As shown in FIG. **5A**, the third rod part **3211** on the third end **321** of the inner scissor frame **32** is rotatably connected with the rotary connection parts **121** of the keycap **10**. In the production process of the long-type key structure **1**, the number of the rotary connection parts **121** may be adjusted according to the practical requirements. For example, the number of the rotary connection parts **121** is two or more than two. In addition, the fourth rod part **3221** on the fourth end **322** of the inner scissor frame **32** are movably connected with the base connection parts **41** of the base plate **40**.

[0039] As shown in FIG. **5B**, the first rod part **3111** on the first end **311** of the outer scissor frame **31** is movably connected with the movable connection parts **122** of the keycap **10**. In addition, the second rod part **3121** on the second end **312** of the outer scissor frame **31** is movably connected with the base connection parts **42** of the base plate **40**. Due to the connection between the keycap **10** and the supporting element **30**, the keycap **10** can be moved upwardly or downwardly relative to the base plate **40**.

[0040] In this embodiment, the third rod part **3211** on the third end **321** of each inner scissor frame **32** is pivotally coupled to three rotary connection parts **121** of the keycap **10**. Since three rotary connection parts **121** and the third rod part **3211** are pivotally coupled to each other, the pressing feel on the corner of the keycap **10** will be further enhanced.

[0041] Please refer to FIGS. **6A**, **6B** and **6C**. FIG. **6A** is a schematic cutaway view illustrating the installation of the metallic reinforcement bar in the long-type key structure of the present invention. FIG. **6B** is a schematic side view illustrating the long side of the keycap in the long-type key structure of the present invention. FIG. **6C** is a schematic side view illustrating the short side of the keycap in the long-type key structure of the present invention.

[0042] As shown in FIG. **6A**, the metallic reinforcement bar **20** is pressed into and engaged with the coupling parts **123** and **124** on the bottom surface **12** of the keycap **10**. Consequently, the metallic reinforcement bar **20** is circumferentially arranged beside the long side and the short side of the keycap **10**. Moreover, the portion of the metallic reinforcement bar **20** beside the short side of the keycap **10** is pushed by the raised platform structures **125** at the short side of the keycap **10**. Consequently, the portion of the metallic reinforcement bar **20** beside the short side of the keycap **10** is slightly separated from the bottom surface **12** of the keycap **10**.

[0043] As shown in FIG. **6B**, the distance between the top surface **11** of the long side of the keycap **10** and the metallic reinforcement bar **20** is $h1$. As shown in FIG. **6C**, the distance between the top surface **11** of the short side of the keycap **10** and the metallic reinforcement bar **20** is $h2$. Since the portion of the metallic reinforcement bar **20** beside the short side of the keycap **10** is pushed by the raised platform structures **125**, the distance $h2$ between the top surface **11** of the short side of the

keycap **10** and the metallic reinforcement bar **20** is larger than the distance **h1** between the top surface **11** of the long side of the keycap **10** and the metallic reinforcement bar **20**. Consequently, the four corners of the keycap **10** are slightly higher than the center of the keycap **10**. Due to this structural design, the pressing feel and key triggering sensitivity of pressing the key structure will be enhanced. In addition, the key pressing stroke delay can be effectively reduced.

[0044] Please refer to FIGS. 7A and 7B. FIG. 7A is a schematic perspective view illustrating a portion of the structure of a first exemplary metallic reinforcement bar. FIG. 7B is a schematic perspective view illustrating a portion of the structure of a second exemplary metallic reinforcement bar.

[0045] In the example of FIG. 7A, the shape and the structure of the metallic reinforcement plate **33** match the shape and the structure of the outer scissor frame **31** shown in FIG. 3. Consequently, the metallic reinforcement plate **33** can be flatly covered and combined with the surface of the outer scissor frame **31**.

[0046] In the example of FIG. 7B, the periphery region of the metallic reinforcement plate **33** is additionally equipped with a bent structure **331**. Consequently, the thickness of the metallic reinforcement plate **33** can be reduced, and the weight of the metallic reinforcement plate **33** can be further reduced.

[0047] From the above descriptions, the present invention provides a long-type key structure. Due to the combination of the outer scissor frame and the metallic reinforcement plate, the overall structural strength of the supporting element will be increased. When the key structure is pressed, the pressing force can be effectively transmitted through the supporting element, and the vibration noise generated when the metal moving parts are rotated or swung will be effectively reduced. Furthermore, the third rod part of each inner scissor frame is connected with at least two rotary connection parts of the keycap. Consequently, the pressing feel of pressing the corner of the keycap can be effectively enhanced. Moreover, since the metallic reinforcement bars and the raised platform structures are installed on the long side and the short side of the bottom surface of the keycap, the pressing feel and key triggering sensitivity of pressing the key structure will be enhanced. In addition, the key pressing stroke delay can be effectively reduced. In other words, the technologies of the present invention are industrially valuable.

[0048] While the invention has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the invention needs not be limited to the disclosed embodiments. It is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A long-type key structure, comprising: a keycap; a base plate; and a supporting element comprising: an outer scissor frame comprising a first end and a second end opposed to the first end; at least two inner scissor frames, wherein each of the at least two inner scissor frames comprises a third end and a fourth end opposed to the third end, the at least two inner scissor frames are pivotally coupled to the outer scissor frame and permitted to be swung relative to the outer scissor frame; and a metallic reinforcement plate with a shape and a structure matching the outer scissor frame, wherein the metallic reinforcement plate is combined on a surface of the outer scissor frame, wherein the first end and the third end are connected with the keycap, and the second end and the fourth end are connected with the base plate, so that the keycap is movable upwardly or downwardly relative to the base plate.

2. The long-type key structure according to claim 1, wherein a bottom surface of the keycap is provided with plural rotary connection parts and plural movable connection parts, a first rod part is formed on the first end, a third rod part is formed on the third end, the first rod part is movably

connected with the plural movable connection parts, and the third rod part is rotatably connected with the plural rotary connection parts.

3. The long-type key structure according to claim 2, wherein the third rod part is rotatably connected with at least two rotary connection parts of the plural rotary connection parts.

4. The long-type key structure according to claim 1, wherein the base plate comprises plural base connection parts, a second rod part is formed on the second end, a fourth rod part is formed on the fourth end, and the second rod part and the fourth rod part are movably connected with the base connection parts.

5. The long-type key structure according to claim 1, wherein the long-type key structure further comprises a metallic reinforcement bar, wherein the metallic reinforcement bar is installed on a bottom surface of the keycap and circumferentially arranged beside a long side and a short side of the keycap.

6. The long-type key structure according to claim 5, wherein plural coupling parts are formed on the bottom surface of the keycap and arranged beside the long side and the short side of the keycap, wherein the metallic reinforcement bar is coupled with the plural coupling parts.

7. The long-type key structure according to claim 5, wherein a raised platform structure is formed on the bottom surface of the keycap and arranged beside the short side of the keycap, and a portion of the metallic reinforcement bar beside the short side of the keycap is pushed by the raised platform structure, wherein a distance between a top surface of the short side of the keycap and the metallic reinforcement bar is larger than a distance between the top surface of the long side of the keycap and the metallic reinforcement bar.

8. The long-type key structure according to claim 1, wherein a bent structure is formed on a periphery region of the metallic reinforcement plate, so that a weight of the metallic reinforcement plate is reduced.
